

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: French Pharmaceutical Regulatory Affairs — Document Compliance NLP

Definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Justification

DrBenchmark's 20-task taxonomy spans NER, POS, MCQA, MCC, MLC, intent classification, and STS [Q1, Q18], providing broad biomedical NLU coverage. However, for the regulatory deployment — which centers on NER and STS over SmPC/PIL/CTD genres — the taxonomy lacks any task explicitly framed around regulatory document compliance. No task evaluates STS over drug-leaflet sentence pairs as a compliance-equivalence judgment, and no task targets regulatory document structure (SmPC sections 4.3/4.4/4.8). The closest genre-aligned NER source is QUAERO/EMA, but it is used only for general biomedical NER with UMLS Semantic Groups. The dataset analysis confirms regulatory document genres are absent across NER (except QUAERO EMA) and across all STS tasks, which the user's HIGH priority weighting on IO makes especially consequential.

Strengths

- Broad task variety covering NER, STS, POS, classification, MCQA — provides multiple complementary signal types relevant to compliance NLP [Q1, Q18]
- QUAERO EMA configuration is the closest genre-aligned NER source, derived from EMA drug leaflets with INNs, excipients, posology, and adverse-event content (QUAERO-D1, QUAERO-D2, QUAERO-D8)
- PxCorpus contributes a fine-grained 38-class posology NER scheme that partially overlaps with regulatory entity needs [Q44] (PXCORPUS-D7, PXCORPUS-D8)
- FrenchMedMCQA contains explicit references to the SmPC ('résumé des caractéristiques du produit'), confirming pharmaceutical regulatory awareness in content (FRENCHMEDMCQA-D3)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required test case categories for the deployment: (a) NER over SmPC/PIL/CTD regulatory text targeting INNs, ATC codes, excipients, posology, and contraindication qualifiers; (b) STS over regulatory sentence pairs calibrated for legal equivalence; (c) version-comparison tasks for safety warning text under variation type IA/IB/II frameworks [WEB-6]. The benchmark provides general biomedical NER and STS but no regulatory-genre-specific category.

ID	Evidence
WEB-6	Variation Type IA/IB/II classification framework establishes legal significance of small text changes — regulatory annotators would score differently than clinical/NLP annotators

IO-2: Source taxonomy omits regionally/contextually relevant categories. No task targets regulatory document genres (SmPC, PIL, CTD); QUAERO EMA NER is the only regulatory-adjacent source and is scored on UMLS Semantic Groups, not regulatory ontology [Q34, Q35]. The dataset analysis confirms no STS task uses regulatory text as a compliance equivalence

target (DEFT2020-D21, CLISTER-D11).

ID	Evidence
WEB-2	DiaMED IAA Table 4 Kappa values not surfaced in search results — full paper PDF retrieval required
DEFT2020-D21	- Ne pas utiliser chez les personnes présentant des difficultés de déglutition en raison du risque d'inhalation bronchique et de pneumopathie lipoïde. — Patient-facing leaflet contraindication — not a structured SmPC section 4.3 entry
CLISTER-D11	La patiente a eu, dans le même temps opératoire, une lithotritie balistique du calcul par voie endoscopique permettant de dénuder le DIU puis de l'extraire par une pince à corps étranger. — Surgical narrative typical of clinical case reports, not regulatory documents
QUAERO-D2	TYSABRI 300 mg solution à diluer pour perfusion natalizumab Chaque flacon de 15 ml de concentré contient 300 mg de natalizumab (20 mg / ml) ; phosphate de sodium ... polysorbate 80 (E433) et eau pour préparation injectable — Full SmPC composition section with INN, dosage, excipients

IO-3: Some tasks are largely irrelevant to the regulatory deployment. FrenchMedMCQA exam-question format is categorically distinct from regulatory NLP workflows (FRENCHMEDMCQA-D11, FRENCHMEDMCQA-D16); MORFITT specialty classification of research abstracts has no regulatory genre overlap (MORFITT-D8, MORFITT-D10); ~40–50% of DEFT2020 STS pairs derive from encyclopedic non-medical content (DEFT2020-D11 through DEFT2020-D15) — diluting the relevant signal.

ID	Evidence
FRENCHMED MCQA-D11	La trinitrine: Est le trinitrate d'isosorbide / Est un médicament anti-angoreux / S'utilise par voie orale dans le traitement de la crise d'angor — Exam false-statement discrimination task — didactic format categorically different from regulatory prose
MORFITT-D8	Des rats ont été assignés aléatoirement à 1 de 3 groupes : un groupe témoin (n = 15), un groupe (n = 15) dans lequel la pancréatite aiguë a été induite au moyen de L-arginine... — Experimental animal study genre; entirely absent from regulatory document context

IO-4: Significant content validity gap: the benchmark's task taxonomy underrepresents the regulatory compliance construct (no SmPC/PIL/CTD-genre task; no regulatory-equivalence STS) while including peripheral content (encyclopedic STS pairs, exam questions, veterinary research). Tropical pathology and overseas territory subcategories also absent (MORFITT-D7 incidental only).

ID	Evidence
WEB-2	DiaMED IAA Table 4 Kappa values not surfaced in search results — full paper PDF retrieval required
DEFT2020-D21	- Ne pas utiliser chez les personnes présentant des difficultés de déglutition en raison du risque d'inhalation bronchique et de pneumopathie lipoïde. — Patient-facing leaflet contraindication — not a structured SmPC section 4.3 entry

Information Gaps

- Whether MANTRAGSC fr_emea/fr_patents subsets (not sampled) provide a stronger regulatory genre signal than the analyzed fr_medline split

Requires Expert Verification

- Whether FrenchMedMCQA's pharmacy-exam content includes a sub-domain that could function as a proxy regulatory knowledge probe for the deployment

Remediation

Gap 1: No task targets SmPC, PIL, or CTD modules as compliance-checking genres; no version-comparison task exists

Recommendation: Add SmPC/PIL section-aware NER and STS tasks (e.g., Section 4.3 contraindications, Section 4.4 special warnings, Section 4.8 adverse reactions per the EMA QRD template) and a version-pair STS task evaluating before/after safety-warning revisions under documented variation classifications [WEB-6, WEB-15].